

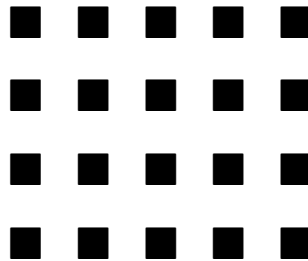
Finding the Missing Factor

Finding a missing factor from an equal-groups picture, from a number sentence, and from a story that names its units

How to hunt for a missing factor.

- A missing factor asks: *how many in each equal group*, or *how many equal groups*.
- Think of the matching division fact. If a number sentence says $\square \times 6 = 30$, then $30 \div 6$ names the number in the box.
- Check your answer by multiplying back. The product must be the total you started with.
- When the problem tells a story, write the unit next to your answer, such as *seats* or *bags*.

1. The tiles below sit in 4 equal rows. There are 20 tiles in all. Write the missing factor in the box, then say how many tiles are in each row.



$$4 \times \underline{\quad} = 20$$

Answer: _____

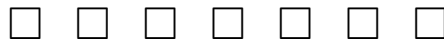
- 2.** Each bag holds the same number of marbles. There are 3 bags and 18 marbles in all. Write the missing factor, then say how many marbles are in one bag.



$$3 \times \underline{\quad} = 18$$

Answer: _____

- 3.** Every row in this seating chart holds 7 seats. The hall has 28 seats in all. The missing factor is the number of rows. Write it in the box.



$$\underline{\quad} \times 7 = 28$$

Answer: _____

4. Find the missing factor. Show the multiplication fact or the division fact you used.

$$\underline{\hspace{1cm}} \times 8 = 56$$

Answer:

5. Find the missing factor by using the division fact that matches this multiplication sentence. Write the division fact you used.

$$9 \times \underline{\hspace{1cm}} = 63$$

Answer:

6. A hall has 42 chairs set out in 6 equal rows. How many chairs are in each row? Write the missing-factor equation you used before you answer.

Answer: chairs

7. A teacher has 45 stickers. She puts exactly 9 stickers in each pack and uses every sticker. How many packs does she fill? Write the missing-factor equation you used before you answer.

Answer: _____ packs

8. This division sentence is missing a number. Find it, then write the multiplication fact that checks your answer.

$$72 \div \underline{\quad} = 8$$

Answer: _____

9. Mia had to find the missing factor in $4 \times \square = 36$. She wrote 32 in the box. Explain in one sentence what Mia did instead of finding a missing factor, then write the correct missing factor.

Answer: _____

10. Ben lays out 56 counters in equal rows with none left over.

(a) He uses 7 rows. How many counters are in each row?

(b) He gathers the same 56 counters and lays them in 8 rows instead. How many counters are in each row now?

Write both missing-factor equations, then explain in one sentence why the number in each row got smaller when the number of rows got bigger.

Answer: _____